

Ice Cream Parlour Management System (ICPMS)

Backend Architecture & API Requirements

Version: 1.0

Technology Stack:

- Node.js
- Express.js
- MongoDB Atlas
- Mongoose
- JWT Authentication
- Bcrypt
- Cloudinary
- Multer
- Express Validator
- Morgan
- Helmet
- CORS
- Dotenv

1. Backend Objectives

The backend must:

- Handle business logic
- Manage authentication
- Manage role permissions
- Process orders
- Generate reports
- Manage inventory
- Manage customers
- Manage employees
- Store product images in Cloudinary
- Provide secure REST APIs

The architecture should be scalable and production-ready.

2. System Architecture

Client (React Frontend)



REST API Layer



Business Logic Layer



MongoDB Database



Cloudinary Storage

3. Backend Folder Structure

backend/

src/

config/

database.js

cloudinary.js

controllers/

auth.controller.js

user.controller.js

product.controller.js

inventory.controller.js

order.controller.js

customer.controller.js

employee.controller.js

offer.controller.js

report.controller.js

upload.controller.js

services/

auth.service.js

product.service.js

inventory.service.js

order.service.js

customer.service.js

employee.service.js

offer.service.js

report.service.js

middlewares/

auth.middleware.js

role.middleware.js

error.middleware.js

validation.middleware.js

upload.middleware.js

validators/

auth.validator.js

product.validator.js

inventory.validator.js

order.validator.js

customer.validator.js

employee.validator.js

routes/

auth.routes.js

user.routes.js

product.routes.js

inventory.routes.js

order.routes.js

customer.routes.js

employee.routes.js

offer.routes.js

report.routes.js

upload.routes.js

models/

User.js

Product.js

Category.js

Inventory.js

Customer.js

Order.js

Invoice.js

Employee.js

Offer.js

Notification.js

utils/

jwt.js

response.js

constants.js

server.js

app.js

.env

4. Authentication System

Authentication Method:

JWT Token Authentication

Password Encryption:

bcrypt

Token Expiry:

7 Days

5. User Roles

Admin

Permissions:

- Full Access
-

Manager

Permissions:

- Products
 - Inventory
 - Orders
 - Reports
-

Cashier

Permissions:

- Orders
 - Billing
 - Customers
-

Inventory Staff

Permissions:

- Inventory
 - Stock Updates
-

6. Authentication APIs

BASE URL

/api/auth

POST /login

Purpose:

User Login

Request:

```
{ "email": "admin@gmail.com", "password": "123456" }
```

Response:

```
{ "success": true, "token": "jwt_token", "user": {} }
```

POST /register

Admin Only

Create Employee/User Account

POST /change-password

Authenticated User

POST /forgot-password

Send Reset Link

POST /reset-password

Reset Password

GET /me

Get Current User

POST /logout

Logout User

7. Product Management APIs

BASE URL

/api/products

GET /

Get All Products

Supports:

- Search
 - Pagination
 - Sorting
 - Filtering
-

GET /:id

Get Product Details

POST /

Create Product

Admin

Manager

PUT /:id

Update Product

DELETE /:id

Delete Product

PATCH /status/:id

Activate / Deactivate Product

8. Product Image Upload

BASE URL

/api/upload

POST /product-image

Upload To Cloudinary

Request:

multipart/form-data

Response:

```
{ "imageUrl": "cloudinary_url" }
```

Supported:

jpg

png

webp

DELETE /product-image

Delete Cloudinary Image

9. Inventory APIs

BASE URL

/api/inventory

GET /

Get Inventory

GET /:id

Get Inventory Item

POST /

Add Inventory Item

PUT /:id

Update Inventory Item

DELETE /:id

Delete Inventory Item

PATCH /stock-in/:id

Increase Stock

PATCH /stock-out/:id

Decrease Stock

GET /low-stock

Get Low Stock Items

10. Customer APIs

BASE URL

/api/customers

GET /

Get Customers

GET /:id

Customer Details

POST /

Create Customer

PUT /:id

Update Customer

DELETE /:id

Delete Customer

GET /:id/orders

Customer Order History

GET /:id/analytics

Customer Analytics

11. Employee APIs

BASE URL

/api/employees

GET /

All Employees

GET /:id

Employee Details

POST /

Add Employee

PUT /:id

Update Employee

DELETE /:id

Remove Employee

PATCH /attendance/:id

Mark Attendance

GET /performance/:id

Performance Analytics

12. Order APIs

BASE URL

/api/orders

GET /

All Orders

GET /:id

Single Order

POST /

Create Order

Workflow:

Create Order

↓

Calculate Tax

↓

Apply Discount

↓

Generate Invoice

↓

Update Inventory

↓

Save Payment

PUT /:id

Update Order

DELETE /:id

Cancel Order

PATCH /status/:id

Update Status

Statuses:

Pending

Preparing

Ready

Completed

Cancelled

13. Invoice APIs

BASE URL

/api/invoices

GET /

Get Invoices

GET /:id

Invoice Details

POST /generate

Generate Invoice

GET /download/:id

Download PDF

14. Offers APIs

BASE URL

/api/offers

GET /

All Offers

POST /

Create Offer

PUT /:id

Update Offer

DELETE /:id

Delete Offer

PATCH /activate/:id

Activate Offer

PATCH /deactivate/:id

Deactivate Offer

15. Reports APIs

BASE URL

/api/reports

GET /daily-sales

GET /weekly-sales

GET /monthly-sales

GET /inventory

GET /customers

GET /products

GET /revenue

Query Parameters:

startDate

endDate

16. Notification APIs

BASE URL

/api/notifications

GET /

User Notifications

PATCH /read/:id

Mark Read

DELETE /:id

Delete Notification

17. Middleware Architecture

Auth Middleware

Purpose:

Verify JWT Token

Role Middleware

Purpose:

Verify Permissions

Example:

Admin Only

Manager Only

Cashier Only

Validation Middleware

Purpose:

Validate Requests

Error Middleware

Purpose:

Centralized Error Handling

Upload Middleware

Purpose:

Handle File Uploads

18. Validation Rules

Products

Name Required

Price > 0

Category Required

Stock >= 0

Customer

Name Required

Mobile Required

Valid Mobile Format

Order

Minimum One Item

Valid Payment Method

Valid Customer

Employee

Name Required

Email Unique

Role Required

19. Response Format

Success Response

```
{ "success": true, "message": "Operation successful", "data": {} }
```

Error Response

```
{ "success": false, "message": "Error message" }
```

20. Security Requirements

JWT Authentication

Password Hashing

Role Based Access

Helmet Security Headers

Rate Limiting

Input Sanitization

Mongo Injection Prevention

XSS Protection

Secure Environment Variables

CORS Configuration

21. Logging

Morgan Logger

Track:

API Requests

Errors

Response Times

Authentication Events

22. Cloudinary Integration

Purpose:

Store Product Images

Workflow:

Frontend Upload

↓

Backend Receives File

↓

Cloudinary Upload

↓

Store URL In MongoDB

↓

Return URL To Frontend

Product Collection Stores:

imageUrl

publicId

23. Business Logic Rules

Product Deleted



Cannot Delete If Active Orders Exist

Order Created



Inventory Automatically Reduced

Order Cancelled



Inventory Automatically Restored

Low Stock Reached



Generate Notification

Invoice Generated



Save Invoice Record

24. Performance Requirements

API Response:

< 500ms

Pagination:

Required For Large Lists

Database Indexes:

Required On

email

mobile

productName

orderNumber

invoiceNumber

25. Future Scalability

Multi Branch Support

Online Orders

QR Ordering

Mobile App APIs

WhatsApp Integration

Third Party Delivery Integration

Advanced Analytics

AI Sales Forecasting

Backend Deliverables

✓ JWT Authentication

✓ Role Based Access Control

- ✓ Product Management APIs
- ✓ Inventory APIs
- ✓ Customer APIs
- ✓ Employee APIs
- ✓ Orders APIs
- ✓ Invoice APIs
- ✓ Reports APIs
- ✓ Notification APIs
- ✓ Cloudinary Integration
- ✓ MongoDB Integration
- ✓ Security Middleware
- ✓ Production Ready Architecture

End of Document